

第 3 章作业 1: SQL (2026 春)

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一、解答题

1. 使用product-db.sql中的 SQL 语句创建关系数据库 Product-DB, 使用 SQL 完成下列查询。

(a) Find the manufacturers that sell laptops but not PC's. (使用集合差运算)

```
select distinct maker from product where type = 'laptop' except  
select distinct maker from product where type = 'pc';
```

(b) Find the model numbers of all printers that are cheaper than the printer model 3002. (使用内连接查询)

```
select p2.model from  
printer as p1 join  
printer as p2 on  
p1.price > p2.price and p1.model = 3002;
```

(c) Find the PC model with the highest available speed. (使用外连接查询)

```
select pc1.model from  
pc as pc1 left outer join  
pc as pc2 on pc1.speed < pc2.speed  
where pc2.model is null;
```

(d) Find the manufacturers of PC's with at least three different speeds. (使用内连接查询)

```
create view tmp as  
select distinct *  
from product natural join pc;  
select distinct t1.maker  
from tmp as t1  
join tmp as t2 on t1.maker = t2.maker  
join tmp as t3 on t1.maker = t3.maker  
where t1.speed != t2.speed  
and t2.speed != t3.speed
```

(e) Find the manufacturers of PC's with at least three different speeds. (使用分组查询)

```
select maker  
from product natural join pc  
group by maker  
having count(distinct speed) >= 3;
```

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(b) Find the model numbers of all printers that are cheaper than the printer model 3002. (使用内连接查询)

(c) Find the PC model with the highest available speed. (使用外连接查询)

(d) Find the manufacturers of PC's with at least three different speeds. (使用内连接查询)

(e) Find the manufacturers of PC's with at least three different speeds. (使用分组查询)